Tahmidul Bin Ferdous

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Education

BSc in Computer Science & Engineering *2022 – 2026*
International Islamic University Chittagong (IIUC), Chittagong, Bangladesh
Undergraduate Thesis: "Detecting Data Leakage in Lung Cancer CT Classification, across CNN, EfficientNet, and Vision Transformer models."
GPA: 3.31 / 4.00

Publications

- *Manuscripts under preparation.*

Technical Skills

- **Programming Languages:** Python (Primary, 250+ logged experiments), C++, SQL, Bash/Shell, LaTeX.
- **Deep Learning Frameworks:** PyTorch (Advanced), JAX/Flax, torchvision, timm, TensorFlow/Keras, Hugging Face.
- **Computer Vision & ML Libraries:** OpenCV, MediaPipe, Ultralytics YOLOv11, scikit-learn, Albumentations, NumPy, Pandas, Matplotlib.
- **Systems & Infrastructure:** Distributed Data Parallel (`torchrun`, multi-GPU DDP), Google Cloud TPU v5e (8-device JAX), Mixed Precision (AMP), Weights & Biases (250 tracked runs), Git, Docker, Linux/Ubuntu.
- **Models Built from Scratch:** Autoregressive Decoder Transformer (GPT-1), Rectified Flow / Flow Matching with ODE solvers, Conditional DDPM, WGAN-GP (gradient penalty & critic loop), DCGAN, Mamba-Attention hybrid (DSMaT).

Research & Professional Experience

EliteLab *Remote*
Student Researcher *2026 – Present*

Outlier AI *Remote*
AI Data Quality & Alignment Specialist *November 2024 – February 2025*

- Audited and curated large-scale human preference and reasoning datasets utilized for Reinforcement Learning from Human Feedback (RLHF) and alignment.
- Refined multi-turn annotation guidelines, directly mitigating prompt ambiguity and improving inter-annotator reliability across reasoning benchmarks.

Personal Projects

Data Leakage in Lung Cancer CT Classification (Undergraduate Thesis) *Sep 2025 – Jun 2026*
PyTorch, timm, CCAL, EfficientNet B0–B4, ViT-B/16, Weights & Biases (188 runs) | *Code*

- Audited slice-level data splits on the IQ-OTH/NCCD lung CT dataset across four architecture families (CCAL dual-CNN attention, EfficientNet B0–B4, ViT-B/16, and a PLOS ONE four-block CNN), training every model twice under a shuffled and a file-order split with everything else held fixed.
- Showed that shuffled slice splits inflate test accuracy by $\sim$**7.5 points** at a matched split ratio over three seeds (CCAL 95.8% $\rightarrow$ 88.3%; EfficientNet 99.3% $\rightarrow$ 91.7%).

Conditional Rectified Flow for CT Synthesis on Google Cloud TPU v5e *Feb 2026 – Mar 2026*
PyTorch, JAX/Flax, Google Cloud TPU v5e, Weights & Biases (18 runs, 32.1 GPU-hours) | *Code*

- Implemented class-conditional Rectified Flow (flow matching) from original mathematical literature: linear interpolant trajectory, velocity-prediction objective, logit-normal timestep sampling, and Euler/midpoint ODE numerical solvers.
- Built multi-GPU DistributedDataParallel (DDP) PyTorch training infrastructure, then **ported the complete architecture to JAX/Flax for 8-device Google TPU v5e**, achieving **576 images/sec** (2.50 s/epoch vs. 15.5 s/epoch on 2$\times$ Tesla T4)—a $\sim$**6**$\times$ **throughput speedup**.

BanglaPoemGPT — 10M-Parameter Autoregressive Transformer from Scratch *March 2025*
Python, PyTorch, Kaggle P100 (4 runs, 2.0 GPU-hours) | *Code*

- Engineered a GPT-1 architecture from first mathematical principles without high-level library wrappers: multi-head causal self-attention, learned positional embeddings, custom tokenizer, and autoregressive generation loop.
- Trained on a 3M-token Bengali poetic corpus, achieving validation cross-entropy loss of **1.693** with coherent generative sampling.

Conditional DDPM for Minority-Class CT Balancing *February 2026*
PyTorch, DistributedDataParallel, 2$\times$ Tesla T4 (5 runs, 4.2 GPU-hours) | *Code*

- Constructed a class-conditional Denoising Diffusion Probabilistic Model (DDPM) featuring a conditional U-Net backbone, cosine noise schedule, and multi-GPU DDP orchestration, reaching a training loss of **0.0238** over 3,000 epochs.

Edge Vision: YOLOv11 Campus Security & Facial Biometric Systems *Oct 2024 – Nov 2024*

YOLOv11, OpenCV, FaceNet, PyTorch | *Code, Kaggle* | *Demos: ID Card Detection, Facial Attendance*

- Curated, annotated, and trained custom YOLOv11-s/m models for campus ID compliance monitoring, reaching **mAP50 = 0.82**.
- Implemented edge facial recognition utilizing SSD face detectors and FaceNet 128-d embeddings with cosine similarity matching, calibrating decision thresholds against false-accept/false-reject trade-offs.

Generative Adversarial Networks: Bengali Digits & Faces *Nov 2024*

PyTorch, DCGAN, WGAN-GP, Weights & Biases (42 runs) | *Code (Digits), Code (Faces)*

- Implemented DCGAN and WGAN-GP (gradient penalty, critic loop) from scratch to generate handwritten Bengali digits and face images, comparing training stability across adversarial objectives.

Sketch AI: Real-Time Air Canvas *Nov 2024*

MediaPipe, OpenCV, Python | *Demo*

- Built a real-time air-canvas that tracks fingertip landmarks from a webcam to draw on screen with hand gestures.

Face Mouse: Hands-Free Cursor Control *Aug 2024*

MediaPipe, OpenCV, PyAutoGUI | *Demo*

- Built a hands-free mouse for users with motor disabilities: nose landmarks drive the cursor and eye blinks trigger clicks.

Certifications

- **DeepLearning.AI Specializations (Coursera):** Neural Networks & Deep Learning, Improving Deep Neural Networks (Tuning, Regularization, Optimization), Convolutional Neural Networks, Sequence Models.

Academic References

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